



DYNAMICS & CONTROL 3

CONTROL

Section:

Tutorials

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Dynamics and Control (UESTC 3001)

- Lecture 1: Root Locus Analysis
- Lecture 2: Root Locus II and Nyquist Plots
- Lecture 3: Bode Plots
- Lecture 4: Bode Plots II
- Lecture 5: Stability in Frequency Domain
- Lecture 6: Stability cont. and Stability Examples
- Lecture 7: Compensators
- **Lecture 8: Tutorials and Test Exercises**

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Notes prepared by: Dr Ola R. Popoola

Tutorial Session 1 (16:25-16:55)

Attempt the question

Can work in groups

Can discuss solutions

Tutorial Session (16:25 – 16:55)

Q. Consider the unity feedback control system with an open loop transfer function of

$$G(s) = \frac{K}{s(s + 5)(s + 10)}$$

- a. Construct the root locus of the system and determine the value of K for which the system is marginally stable.
- b. Sketch the bode magnitude and phase plots of the system.
- c. Using the Nyquist criterion, find the range of values of K for which the system is stable

A. Root Locus:

Method to plot the root locus:

- 1- Find the number of asymptotes by identifying the poles and zeros of the transfer function;
- 2- Calculate the centre of gravity;
- 3- Calculate the angle of the asymptotes;
- 4- Find the breakaway point;
- 5- Plot the root locus.

B. Bode Plot:

Steps to sketch the bode plot:

1. Write transfer function in time constant form and identify the components
2. Identify corner frequencies and the magnitude and phase behaviour of components
3. Plot the Bode magnitude plot with magnitude in dB vs $\log(\omega)$
4. Plot the Bode phase plot with phase in deg vs $\log(\omega)$

C. Point/Region of instability:

Steps to find point of instability using Nyquist criterion:

1. Write $G(j\omega)$ and rationalise to get real part and imaginary parts
2. Equate the imaginary part to zero and find the value of ω_c where the Nyquist plot cuts the negative real axis
3. Put the value of ω_c into $G(j\omega)$ to get the range of K for stability where $G(j\omega_c) < 1$ or use 4
4. Write the magnitude of $G(j\omega)$ compute for $G(j\omega_c)$. The get the range of K for $G(j\omega_c) < 1$

1 – Find the number of asymptotes by identifying the poles and zeros of the transfer function

$$G(s) = \frac{K}{s(s+5)(s+10)}$$

$$\begin{aligned}\text{Number of Asymptotes} &= \text{Num of Poles} - \text{Num of Zeros} \\ &= 3 - 0 = 3\end{aligned}$$

Poles are at: 0, -5, -10

There will be locus between 0 and -5, none between -5 and -10, locus between -10 and -inf

2- Calculate the centre of gravity

$$G(s) = \frac{K}{s(s+5)(s+10)}$$

$$\sigma = \frac{\sum poles - \sum zeros}{Num_{poles} - Num_{zeros}}$$

$$\sigma = \frac{0 + (-5) + (-10)}{3 - 0} = -\frac{15}{3} = -5$$

3- Calculate the angle of the asymptotes

$$G(s) = \frac{K}{s(s+5)(s+10)}$$

$$\theta = \frac{(2q+1) * 180^\circ}{Num_{poles} - Num_{zeros}} ; q = 0,1,2$$

$$\theta = \frac{([1, 3, 5]) * 180}{3 - 0} = \frac{[180, 540, 900]}{3}$$

$$\theta = [60, 180, 300]$$

4- Find the breakaway points

$$G(s) = \frac{K}{s(s+5)(s+10)}$$

$$1 + G(s)H(s) = 0$$

$$1 + \frac{K}{s(s+5)(s+10)} = 0$$

$$s(s+5)(s+10) + K = 0$$

$$K = -s^3 - 15s^2 - 50s$$

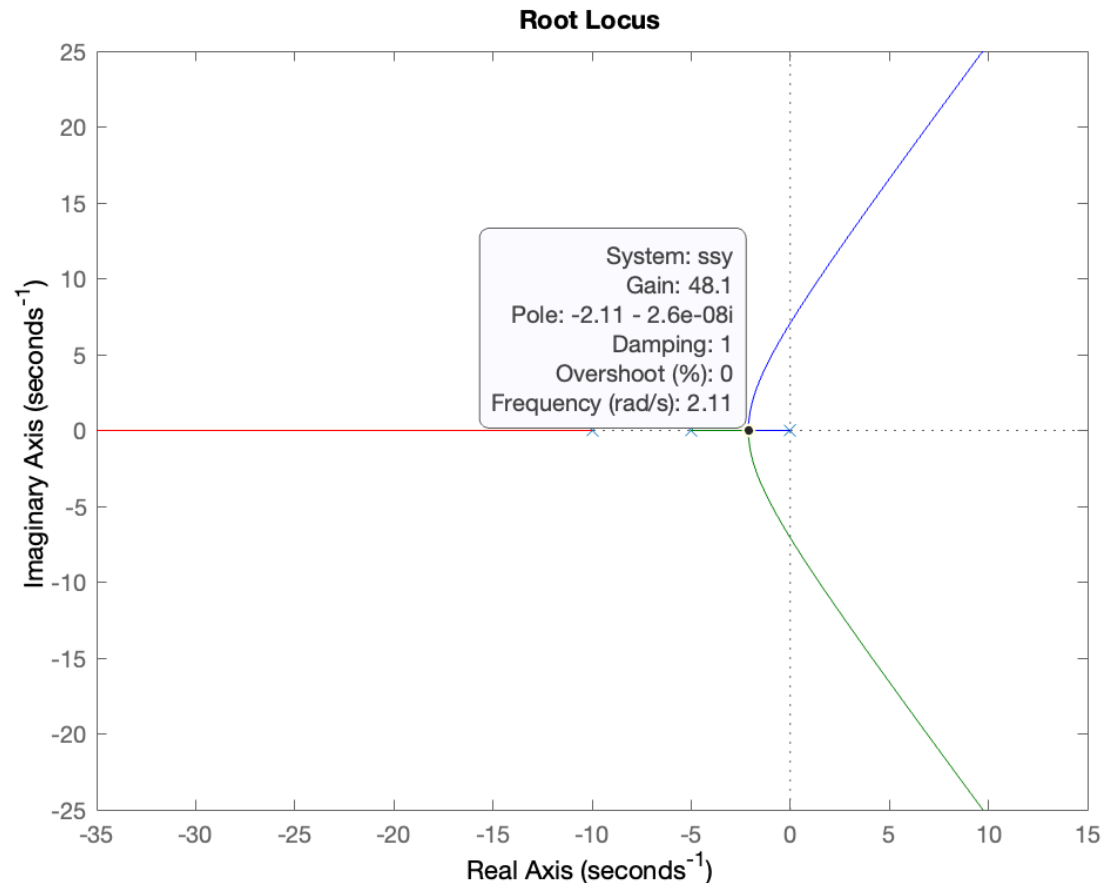
$$\frac{dK}{ds} = 0; \left(\frac{d(-s^3 - 15s^2 - 50s)}{ds} \right) = 0$$

$$-3s^2 - 30s - 50 = 0$$

$$s = -2.113, -7.88$$

5- Sketch the root locus

$$G(s) = \frac{K}{s(s+5)(s+10)}$$



Value of K for marginal stability

$$s^3 + 15s^2 + 50s + K = 0$$

s^3	1	50
s^2	15	K
s^1	$\frac{750 - K}{15}$	K
s^0	K	

K for marginal stability = 750

For frequency of marginal stability or sustained oscillations

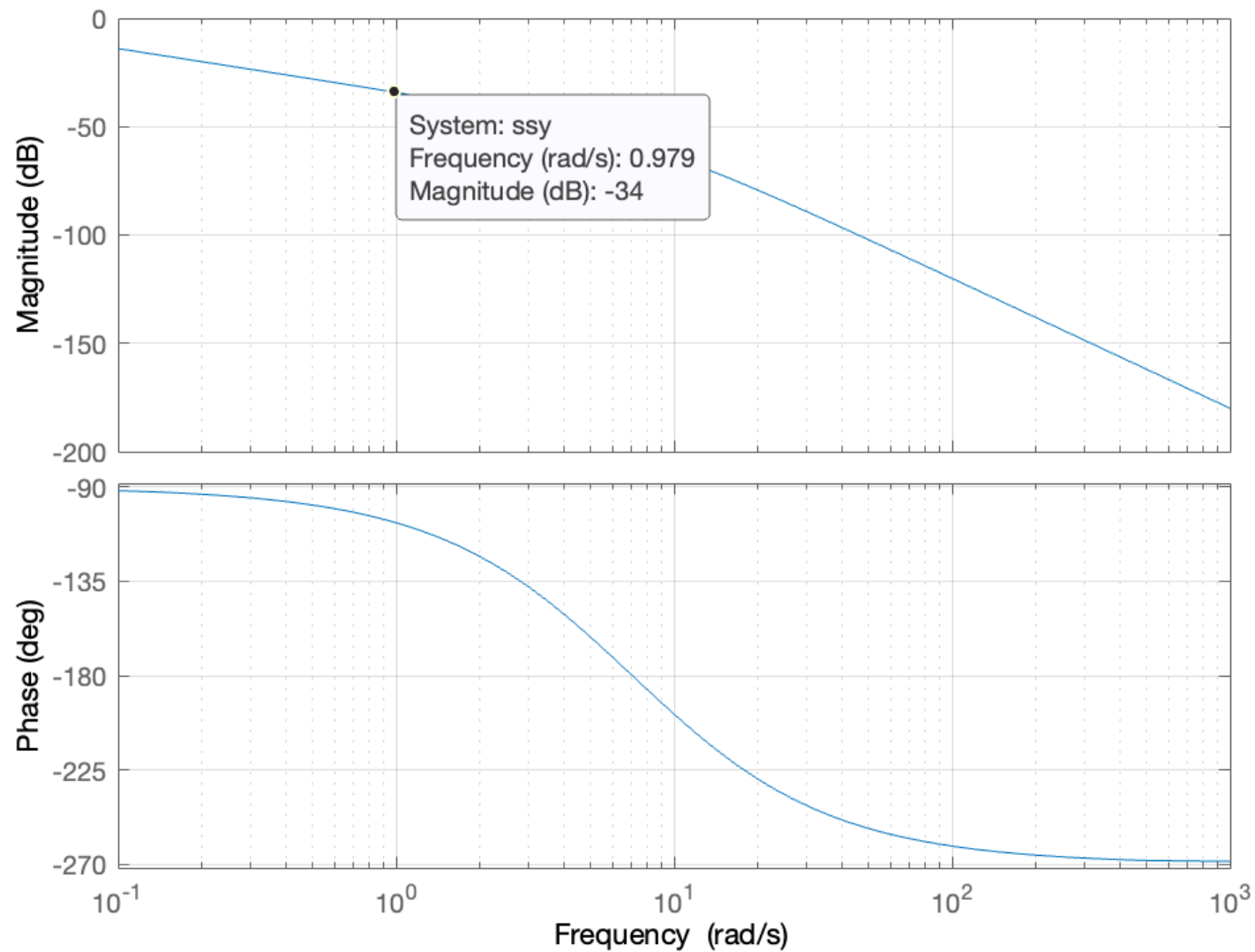
$$15s^2 + K = 0; 15s^2 + 750 = 0; s = \pm j7.07$$

$$G(s) = \frac{K}{s(s+5)(s+10)} = \frac{K}{50s \left(\frac{s}{5} + 1\right) \left(\frac{s}{10} + 1\right)}$$

Component	Corner Frequency	Asymptotic log-Magnitude Characteristics	Phase angle characteristic
$\frac{0.02}{j\omega}$	<i>None</i>	Slope -20dB/dec, passes $20\log 0.02 = -34\text{dB}$ at $\omega=1$	Constant -90deg
$\frac{1}{1 + \frac{j\omega}{5}}$	$\omega_1 = 5$	0 dB $\omega < \omega_1$ and -20dB/dec $\omega > \omega_1$	Varies from 0 deg to -90 deg angle at $\omega_1 = -45$ deg
$\frac{1}{1 + \frac{j\omega}{10}}$	$\omega_2 = 10$	0 dB $\omega < \omega_2$ and -20dB/dec $\omega > \omega_2$	Varies from 0 deg to -90 deg angle at $\omega_2 = -45$ deg

$$G(s) = \frac{K}{s(s+5)(s+10)} = \frac{K}{50s \left(\frac{s}{5} + 1\right) \left(\frac{s}{10} + 1\right)}$$

Bode Diagram



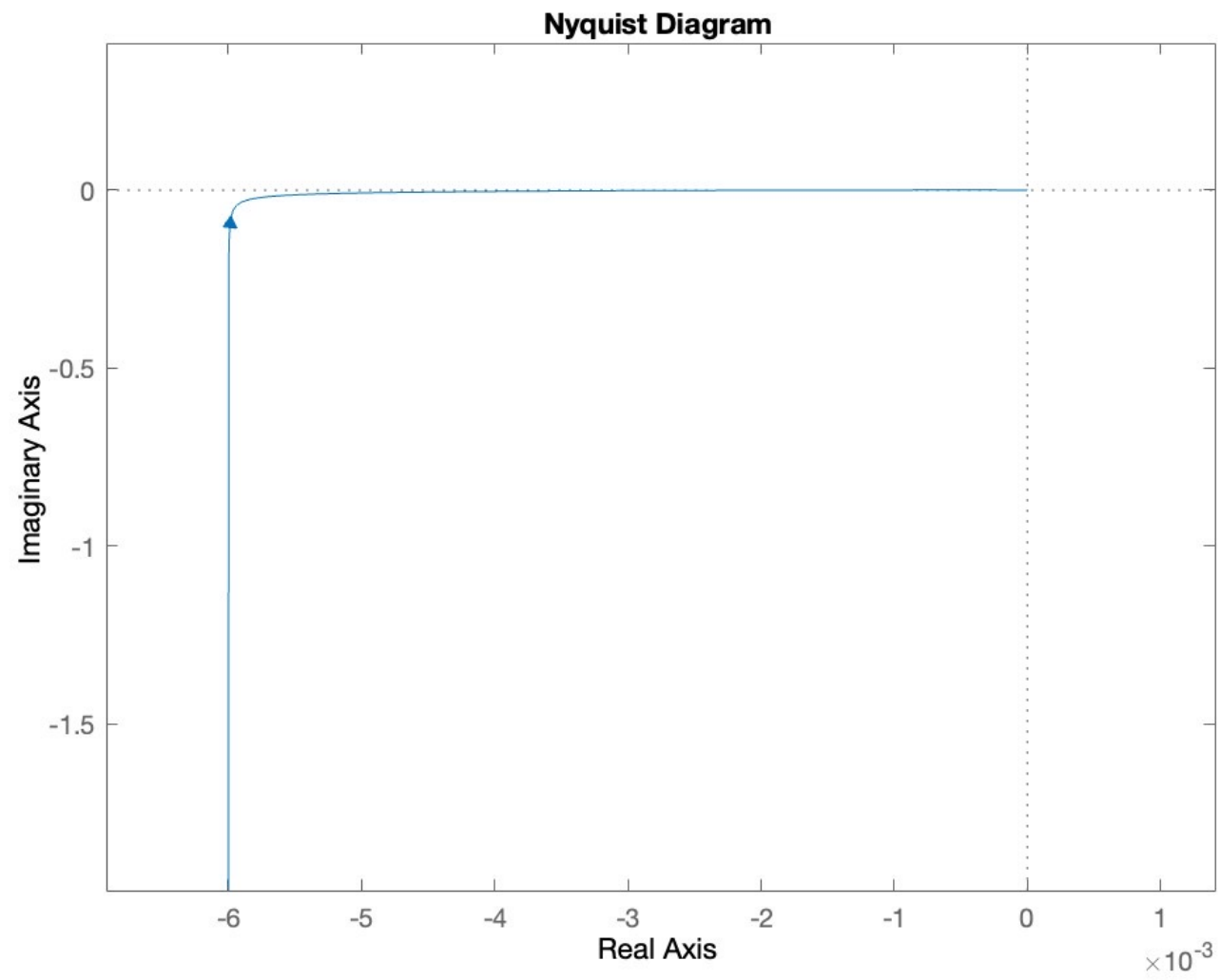
Bonus: Sketch of Nyquist Plot:

$$G(j\omega) = \frac{K}{j\omega(j\omega + 5)(j\omega + 10)} = \frac{K}{-15\omega^2 + j(50\omega - \omega^3)}$$

$$|G(j\omega)| = \frac{1}{\sqrt{225\omega^4 + (50\omega - \omega^3)^2}}$$

$$\text{Arg}(G(j\omega)) = \tan^{-1} \frac{50\omega - \omega^3}{15\omega^2}$$

	Magnitude	Phase
$\omega = 0$	∞	90
$\omega = 10$	0.0006	-18.43
$\omega = \infty$	0	-90



1 – Write $G(j\omega)$ and rationalise to get real part and imaginary parts

$$\begin{aligned}
 G(s) &= \frac{K}{s(s+5)(s+10)} \\
 G(j\omega) &= \frac{K}{j\omega(j\omega+5)(j\omega+10)} = \frac{K}{-j\omega^3 - 15\omega^2 + j50\omega} \\
 &= \frac{K}{-15\omega^2 + j(50\omega - \omega^3)} \\
 &= \frac{-15\omega^2 K}{(-15\omega^2)^2 + (50\omega - \omega^3)^2} - \frac{j(50\omega - \omega^3)}{(-15\omega^2)^2 + (50\omega - \omega^3)^2}
 \end{aligned}$$

2—Equate the imaginary part to zero and find the value of ω_c where the Nyquist plot cuts the negative real axis

$$\frac{j(50\omega - \omega^3)}{(-15\omega^2)^2 + (50\omega - \omega^3)^2} = 0$$

$$50\omega - \omega^3 = 0$$

$$50\omega = \omega^3$$

$$\omega_c = \sqrt{50}$$

3– Put the value of ω_c into $G(j\omega)$ to get the range of K for stability where $G(j\omega_c) < 1$ or use 4

$$G(j\omega) = \frac{-15\omega^2 K}{(-15\omega^2)^2 + (50\omega - \omega^3)^2} - \frac{j(50\omega - \omega^3)}{(-15\omega^2)^2 + (50\omega - \omega^3)^2}$$

$$G(j\omega_c) = \frac{-15\omega_c^2 K}{(-15\omega_c^2)^2 + (50\omega_c - \omega_c^3)^2} - \frac{j(50\omega_c - \omega_c^3)}{(-15\omega_c^2)^2 + (50\omega_c - \omega_c^3)^2}$$

$$G(j\omega_c) = \frac{-15\sqrt{50}^2 K}{(-15.50)^2 + (50\sqrt{50} - \sqrt{50}^3)^2} - \frac{j(50\sqrt{50} - \sqrt{50}^3)}{(-15\sqrt{50}^2)^2 + (50\sqrt{50} - \sqrt{50}^3)^2}$$

$$G(j\omega_c) = \frac{-15\sqrt{50}^2 K}{(-15.50)^2 + (50\sqrt{50} - \sqrt{50}^3)^2} < 1$$

$$K < 750$$

4– Write the magnitude of $G(j\omega)$ compute for $G(j\omega_c)$. Then get the range of K for $G(j\omega_c) < 1$

$$G(j\omega) = \frac{K}{j\omega(j\omega + 5)(j\omega + 10)}$$

$$|G(j\omega)| = \frac{K}{\omega \sqrt{(\omega^2 + 5^2)} \sqrt{(\omega^2 + 10^2)}}$$

$$|G(j\omega_c)| = \frac{K}{\omega_c \sqrt{(\omega_c^2 + 5^2)} \sqrt{(\omega_c^2 + 10^2)}} < 1$$

$$\frac{K}{\sqrt{50} \sqrt{(50 + 5^2)} \sqrt{(50 + 10^2)}} < 1$$

$$\frac{K}{\sqrt{562500}} < 1$$

$$K < 750$$